

COS 433 Problem Set 3

Due date: Wednesday, 10/28/2026 at 11:59pm. See [the course website](#) for our submission and grading policies.

Proof citations. Your proofs may assume without proof any fact stated in lecture or stated on the problem sets. All other statements – including results from textbooks or other sources that were not mentioned in lecture – should be proved in your problem set write-up.

Grading Policy. A total score of 100 points constitutes full credit for the problem set. You are not expected to complete all of the problems. Feel free to work on whichever problems you prefer!

Problem 0: Sign up for Check-in 2

Please use this sign-up link: <https://calendar.app.google/3Bjqzw1pfBAeNzL9A>

Problem 1: CCA-Secure or not? (25 points)

Let $SKE = (Enc, Dec)$ be a CPA-secure encryption scheme, and let $MAC = (Sign, Verify)$ be a MAC. In class, we saw that the encryption scheme given by

$$Enc'((k_1, k_2), m) = (ct \leftarrow Enc(k_1, m), Sign(k_2, ct))$$

and

$$Dec'((k_1, k_2), (ct, \sigma)) = Dec(k_1, ct) \text{ if } Verify(k_2, ct, \sigma) = 1$$

is CCA-secure.

In this problem, we consider other candidate constructions of CCA-secure encryption. Are these constructions secure? Prove or disprove via counterexample.

(a) Define

$$Enc_1((k_1, k_2), m) = Enc\left(k_1, (m, Sign(k_2, m))\right),$$

and define $Dec_1((k_1, k_2), ct)$ to be the following procedure:

- Run $Dec(k_1, ct)$ to obtain a string (m, σ) .
- Output m if $Verify(k_2, m, \sigma) = 1$ and $\perp$ otherwise.

(b) Define

$$\text{Enc}_2((k_1, k_2), m) = \left(\text{Enc}(k_1, m), \text{Sign}(k_2, m) \right),$$

and

$$\text{Dec}_2((k_1, k_2), (\text{ct}, \sigma)) = \text{Dec}(k_1, \text{ct}) \text{ if } \text{Verify}(k_2, \text{Dec}(k_1, \text{ct}), \sigma) = 1.$$

Problem 2: Bit-by-Bit Encryption (15 points)

Let $\text{SKE} = (\text{Enc}, \text{Dec})$ denote a secret-key encryption scheme with key space $\{0, 1\}^\lambda$ and message space $\{0, 1\}$ (one-bit messages). Given such a scheme, we can define an encryption scheme for arbitrary-length messages by encrypting bit-by-bit, i.e.,

$$\text{Enc}(k, m_1 \dots m_n) = \text{Enc}(k, m_1), \dots, \text{Enc}(k, m_n).$$

Prove that the resulting encryption scheme is *never* CCA-secure.

Problem 3: One-Way Functions (30 points)

Suppose that $f = \{f_\lambda : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda\}_{\lambda \in \mathbb{N}}$ is a length-preserving one-way function.

- (a) Is the function $x \mapsto f(f(x))$ also one-way? Prove or disprove. (10 points)
- (b) Is the function $(x_0, x_1) \mapsto f(x_0 \oplus x_1)$ also one-way? Prove or disprove. (10 points)
- (c) Is the function $(x_0, x_1) \mapsto f(x_0) \oplus f(x_1)$ also one-way? Prove or disprove. (10 points)

Problem 4: Square roots mod p (30 points)

In class, we gave an efficient algorithm for computing square roots mod p for all $p \equiv 3 \pmod{4}$. In this problem, we will give an efficient algorithm that works for all primes p .

- (a) Suppose that $p - 1$ is divisible by 2^ℓ but not $2^{\ell+1}$. Give an efficient algorithm that on input (p, a) , if a is a square mod p , finds an integer x such that

$$x^{2^\ell} \equiv a^{2^{\ell-1}} \pmod{p}.$$

Deduce that to compute square roots mod p , it suffices to give an efficient algorithm for computing square roots of a given that $a^{2^{\ell-1}} \equiv 1 \pmod{p}$.

Hint: use Fermat's little theorem.

- (b) For this next part, we will make use of an important fact about the multiplicative group $\mathbb{Z}_p^\times = \{1, 2, \dots, p-1\}$.

Fact 1. There exists a “generator” $g \in \mathbb{Z}_p^\times$ such that $\{1, g, g^2, g^3, \dots, g^{p-2} \pmod{p}\} = \mathbb{Z}_p^\times$.
That is, the powers of g generate $\mathbb{Z}_p^\times$.

Using this fact, prove that there exists an integer ζ with the following properties:

- $\zeta^{2^\ell} \equiv 1 \pmod{p}$.
- For all a with $a^{2^{\ell-1}} \equiv 1 \pmod{p}$, there exists an integer e such that $\zeta^{2^e} \equiv a \pmod{p}$.

Furthermore, give a randomized polynomial-time algorithm for finding such a ζ .

Hint: as a thought experiment, write everything as a power of $g \pmod{p}$.

(c) Finally, give a polynomial time algorithm that, when given a and ζ as above, finds an exponent e such that $\zeta^{2^e} \equiv a \pmod{p}$.

Hint: Find e bit-by-bit.

Problem 5: PKE based on Factoring (30 points)

In class, we showed how to construct a public-key encryption scheme under the Quadratic Residuosity assumption. In this problem, we will see how to construct a PKE scheme under the *weaker assumption* that factoring is hard.

(a) Let $N = pq$ be a Blum integer. Let $\mathbb{QR}_N \subset \mathbb{Z}_N^\times$ denote the set of integers modulo N that are (1) relatively prime to N , and (2) quadratic residues (perfect squares) mod N .

Prove that the function $f_N : \mathbb{QR}_N \rightarrow \mathbb{QR}_N$ given by $f_N(x) = x^2$ is a permutation.

(b) Show that given the factorization of N , it is possible to efficiently invert the permutation f_N .

(c) Let $b(x)$ denote a hard-core predicate for f_N (for simplicity, assume that a deterministic hard-core predicate exists). Construct a public-key encryption scheme using $f_N(\cdot)$, $b(\cdot)$ and prove that it is secure assuming the hardness of factoring.

Hint: think about the PKE construction from class.

(d) **Bonus (challenging):** prove that $b(x) = \text{lsb}(x)$, the least significant bit of x , is a hard-core bit for f_N .

Problem 6: PKE and OWFs (15 points)

Let $(\text{Gen}, \text{Enc}, \text{Dec})$ denote a public-key encryption scheme. Describe (in terms of this scheme) an efficiently computable function f such that if $(\text{Gen}, \text{Enc}, \text{Dec})$ is secure, then f *must* be a one-way function. Prove this conditional one-wayness property of f .

Problem 7: One-wayness vs. Pseudorandomness (25 points)

(a) Let $G = \{G_\lambda : \{0, 1\}^\lambda \rightarrow \{0, 1\}^{2\lambda}\}_\lambda$ be a PRG. Is G necessarily a one-way function? Prove or disprove. (10 points)

(b) Let $G = \{G_\lambda : \{0, 1\}^\lambda \rightarrow \{0, 1\}^{\lambda+1}\}_\lambda$ be a PRG. Is G necessarily a one-way function? Prove or disprove. (15 points)